

Geo-Spatial PM 2.5 Trend Analysis: Seasonal and Yearly Assessment Using GIS and Data Analytics

A Training Internship Report

SUBMITTED BY

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Lastly, I would like to thank my academic institution, Delhi Technological University (DTU) , and my professors for their continuous support and for encouraging me to pursue this opportunity.

This internship has significantly contributed to my development and inspired me to further explore into core environmental engineering. I look forward to applying the knowledge and skills I have gained to future endeavors.

I express my gratitude to my academic advisor **Prof. Sh. A.K. Haritash** for providing his constant encouragement and motivation in the successful completion of this internship and report. Finally, I would like to thank my family for their constant support and encouragement throughout this period.

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About IMD (India Meteorological Department)

The India Meteorological Department (IMD) is the national agency responsible for meteorology, weather forecasting, seismology, and atmospheric observations in India. Established in 1875, it functions under the administrative control of the Ministry of Earth Sciences, Government of India. Its headquarters is located at Mausam Bhawan, Lodhi Road, New Delhi – 110003. IMD also has six regional meteorological centers in Chennai, Mumbai, Kolkata, Guwahati, Nagpur, and New Delhi, along with multiple state centers and field offices across the country.

IMD is entrusted with monitoring weather systems and climate patterns and providing timely and accurate forecasts to protect life and property. It plays a vital role in issuing warnings for extreme weather events such as cyclones, floods, heatwaves, cold waves, and thunderstorms, thereby assisting disaster preparedness and response efforts at national and state levels.

A key function of the department is the provision of public weather services, which include daily weather forecasts, monsoon updates, agricultural advisories, aviation and marine forecasts, and climate summaries. IMD supports agriculture through its agrometeorological services, helping farmers make informed decisions regarding sowing, irrigation, and harvesting. It also serves the needs of sectors such as civil aviation, defense, shipping, water resource management, tourism, and power distribution.

IMD operates a vast observational network comprising over 750 surface observatories, 200 agro-meteorological stations, 43 Doppler Weather Radars (DWRs), and various upper-air, radiation, and satellite monitoring stations. It utilizes satellite data, high-resolution models, and advanced numerical weather prediction (NWP) techniques for real-time analysis and forecasting. The department has also introduced public outreach platforms such as the Mausam App and Meghdoot App for weather information and farmer advisories, respectively.

Introduction

Air pollution is one of the most serious environmental and health issues faced by the world today. Among the many pollutants in the air, **Particulate Matter 2.5 (PM 2.5)** is considered one of the most harmful. These are very fine particles, smaller than 2.5 micrometers in diameter, that can easily enter the lungs and even reach the bloodstream. Long-term exposure to high levels of PM 2.5 has been linked to health problems such as breathing issues, heart disease, and shorter life expectancy.

In India, the problem of PM 2.5 is especially serious due to a fast-growing population, increasing traffic, industrial activity, and changing land use. Many cities across the country face different levels of pollution depending on their location and climate. For example, cities in **North India** like Delhi, Lucknow, and Kanpur often record very high PM 2.5 levels, especially during winter. On the other hand, cities in the **South and along the coasts**, such as Bengaluru, Kochi, and Chennai, usually show lower pollution levels. These differences make it important to study PM 2.5 trends both **over time and across regions**.

To better understand these patterns, this project was carried out at the **India Meteorological Department (IMD)** using **GIS (Geographical Information Systems)** and **data analysis tools**. The study focused on examining PM 2.5 levels across more than 100 Indian cities between **2018 and 2024**. The main goals were to:

- Map **yearly and seasonal pollution levels** in different cities,
- Find out which cities and seasons experience the worst air quality,
- Compare pollution across different regions, and
- Study special cases like the **COVID-19 lockdown**, when pollution levels dropped suddenly.

By using maps and data together, this project gives a clear picture of how PM 2.5 changes with the seasons and from year to year. The findings can help in making better decisions to improve air quality, especially by focusing on specific cities and times of the year when pollution is highest.

Methodology

The study was carried out using a combination of **Geographical Information Systems (GIS)** and **data analytics techniques** to analyze the spatial and temporal behavior of **PM 2.5 concentrations** across major Indian cities from **2018 to 2024**.

1. Data Collection

PM 2.5 data for multiple cities across India was sourced from:

- **Central Pollution Control Board (CPCB)** air quality stations

The dataset contained hourly average concentrations of PM 2.5 for every CPCB station between **2018 and 2024**.

Date Time	Benzene	BP	CO	Eth-Benzene	MP-Xylene	NO	NO2	NOx	Ozone	O Xylene	PM2.5	RH	SO2	SR
2018-01	2.28791608	762.668629	0.82647552	2.14267133		8.43082621	18.5733333	15.2663909	40.6007552	1.69237693	78.3272727		14.4580785	93.5023077
2018-02	1.62052117	763.359788	1.19203583	1.79035831		8.77014706	14.3225163	12.9862255	35.1853583	1.50998371	50.9360586		9.40278502	127.712932
2018-03	0.80220408	760.862095	1.32259864	0.63994558		6.35231293	9.07838095	8.51786395	29.4973333	1.05917695	40.6535675		6.0495102	139.514286
2018-04	0.25521739	758.557391	1.73217391	0.4773913		29.4965217	38.8017391	35.5304348	32.0669565	1.42045455	86.6656522		6.86809524	351.907391
2018-05	0.2825	693.708333	1.096	1.115	1.00333333	6.75571429	13.7813333	10.3886667	38.836	0.73166667	19.0026667	37.8966667	4.36933333	156.5775
2018-06	0.53857143	755.851429	1.44142857	8.15285714	0.32142857	12.0185714	19.43	18.4385714	33.4328571	2.19428571	229.441429	54.39	7.64857143	243.365714
2018-07	0	754.985	1.44125	0.14727273	0	12.4441667	20.7808333	19.4991667	40.3518182	0.77090909	46.3433333	52.1383333	7.144	258.878333
2018-08	0	757.29537	0.92064815	0.27407407	0.08981481	13.137037	18.4899074	31.1487037	30.277963	0.28981481	47.9742593	70.9722222	9.15972222	111.685833
2018-09	0.03513644	758.473547	1.00004823	0.21088283	0.13565008	13.1766132	19.4640931	32.6023435	24.9677368	0.56794543	53.0927609	70.9062921	10.9452013	131.962215
2018-10	0.12572308	760.677589	1.07392915	0.35690769	0.34487692	17.4569403	27.5101045	44.9652239	26.5259524	0.65863077	61.691875	73.5861012	18.633766	99.6459077
2018-11	0.19142061	762.174819	0.96683916	0.95272981	0.43490251	15.8111699	27.7815181	43.592312	28.3133426	1.10377437	58.1658993	74.9421031	12.2497547	79.4089276
2018-12	0.22441667	762.974694	0.90526389	0.82473611	0.51326389	14.5255139	26.49301	40.1484722	25.6892211	0.92131944	76.0097734	74.623125	12.0222594	58.9754306
2019-01	0.20011419	764.717311	0.85473258	1.05190865	0.62429038	4.20478049	14.7712175	18.9436039	34.2863592	0.92911909	60.5971122	70.4443961	15.1658569	68.81219
2019-02	0.05037879	763.428182	0.80244318	0.69342803	0.48931818	3.19007576	12.7569318	15.9467992	30.3494129	0.77070076	65.6236416	70.815947	10.0336398	90.8852273
2019-03	0.04304945	761.296827	0.79354167	0.26337912	0.15596154	8.43907015	12.9524051	25.4443836	19.9634478	0.30583791	51.3366154	68.2375962	6.32065097	137.526415
2019-04	0.11245981	759.04209	1.09903537	0.56270096	0.32958199	8.64330134	9.45076775	19.7839155	15.3424598	0.59207395	26.9648953	69.0809003	4.82337134	137.705
2019-05	0.20912451	756.673658	1.1972179	1.01723735	0.65921212	9.01366667	13.3937292	22.2347917	42.5100984	1.24770428	33.7864414	58.4955152	5.13974	151.070097
2019-06	0.16932203	754.948569	0.97804598	0.71694915		8.69326733	18.6395474	19.9159972	43.8411874	0.86665254	27.3533154		4.59495588	159.432347
2019-07	0.17603064	755.926723	0.70815021	0.71437326		8.07315716	14.8459666	22.9184423	23.721784	0.90207521	32.2057402		4.22563319	115.384948
2019-08	0.15090517	756.118822	0.64892086	0.88041667		8.35596264	15.0504598	23.4061207	16.7689157	0.89033046	25.2063506		7.25677725	112.941066
2019-09	0.0635	757.957361	1.31704802	0.57794444		5.72277143	10.8270085	13.3768091	15.031057	0.69136111	31.1642837		5.38851003	97.9432962
2019-10	0.1559175	759.920973	1.11633609	0.22049787		6.28534483	23.8391329	20.7036732	13.5110179	0.63379801	45.202274		7.84025532	72.9200822

2. Data Preprocessing

Before analysis, the raw hourly PM 2.5 data obtained from CPCB stations had to be standardized and made usable. The following steps were performed:

- **Missing Data Handling:**
 - Short gaps were interpolated linearly
 - Long gaps were either excluded or imputed using median values for that station/month
- **Outlier Detection:**
 - Values beyond acceptable physical limits were either corrected or removed
- **Time Aggregation:**

- Hourly → Daily → Monthly → Seasonal → Annual
- Ensured uniformity across cities and years

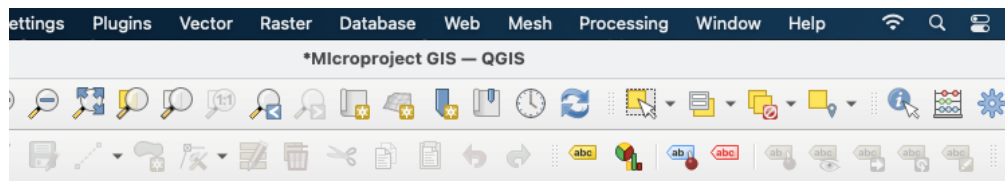
Using Data analysis techniques, the data was preprocessed, cleaned, and finally converted to Subsets of monthly, seasonal, and yearly datasets.

	2018	2019	2020	2021	2022	2023	2024
Agra	101.427694	73.1252895	79.5976194	79.4123451	56.3719055	37.4156702	37.3170496
Ahmedabad	163.101723	61.2989536	47.0084181	53.0006326	45.3268062	50.3398491	42.584499
Aizawl			12.7179063	15.2096056	6.89328401	11.0783835	14.045146
Ajmer	58.3640292	47.7036836	39.803853	45.3787077	51.8124773	46.2920608	41.2599895
Alwar	51.6481794	45.1046884	41.0781701	43.2544752	41.4984303	42.2040702	40.9920392
Amaravati	38.162503	37.650869	28.5185691	28.3689691	29.67192	34.1726739	26.4751333
Ambala		59.790306	57.8066728	61.0154392	68.9984008	43.0595649	43.6078024
Amritsar	42.0480234	50.0630097	46.4388914	50.9563561	50.7535797	59.6375948	57.1184977
Ankleshwar		48.3693115	48.1754149	66.614993	64.0050277	50.2438932	42.966896
Asansol	59.7245781	61.4940993	53.268624	58.7321644	59.1499233	61.2956306	74.8670858
Aurangabad	46.8593839	39.4710823	29.8235092	33.402262	43.4152128	43.150321	57.771432
Bagalkot			29.4724452	20.6151084	24.0901383	23.050418	22.3023332
Baghpat	165.290545	102.04551	106.122655	90.9840701	88.1505488		5.66833333
Bahadurgarh		74.7300412	73.4699397	90.5959153	83.6739699	75.4145365	77.8634604
Ballabgarh		88.1445796	78.9434419	82.0983083	68.4163304	64.9735332	78.0227565

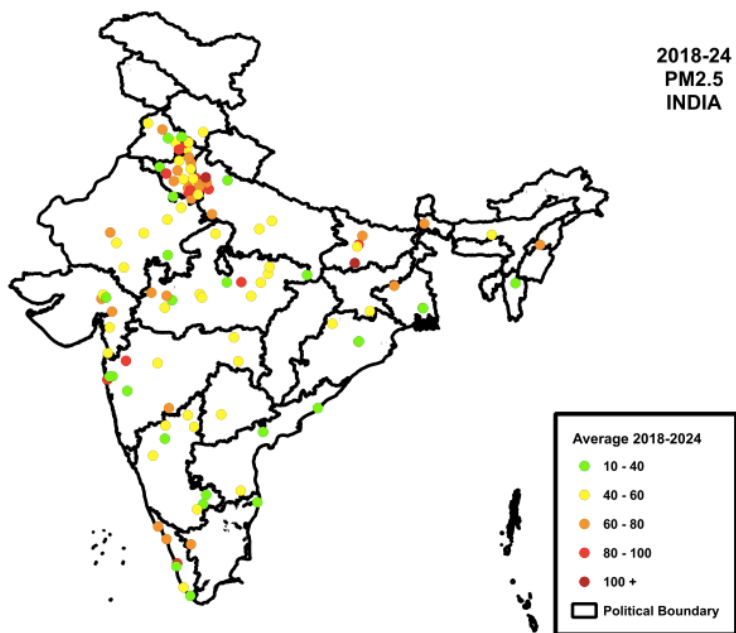
3. Data Visualization using GIS

The processed data was imported into **QGIS**, an open-source GIS tool. The following steps were performed:

- Mapping **PM 2.5 values** onto India's shape file layer
- Creating **thematic color-coded maps** for each year and season
- Using **interpolation techniques** to estimate pollution in surrounding areas



4. Annual Categorization



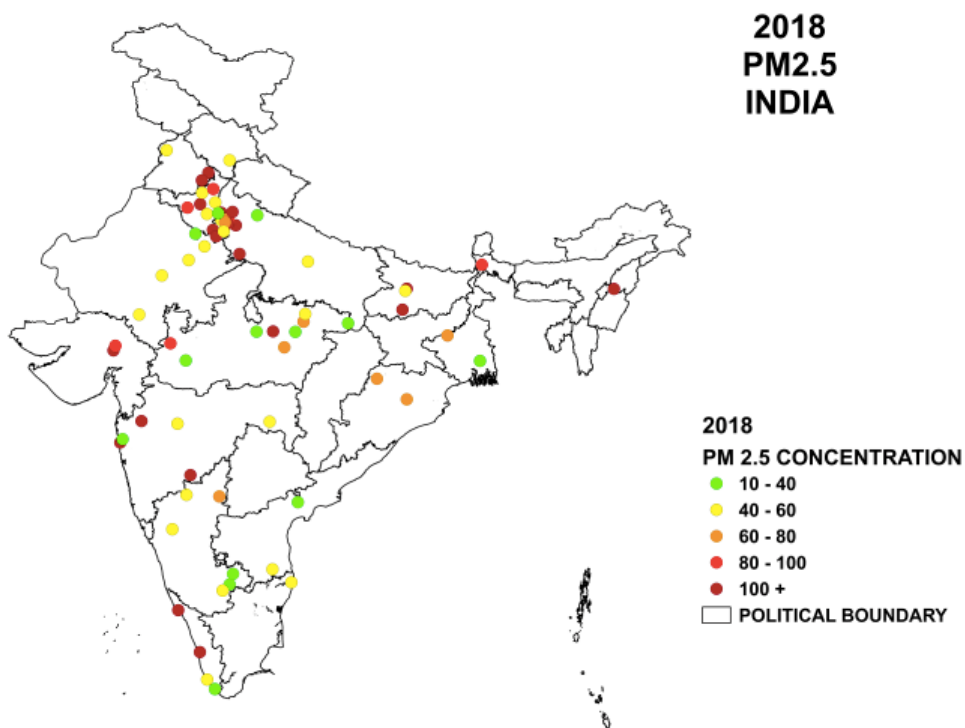
Annual Average PM 2.5 Levels Across Major Indian Cities (2018–2024)

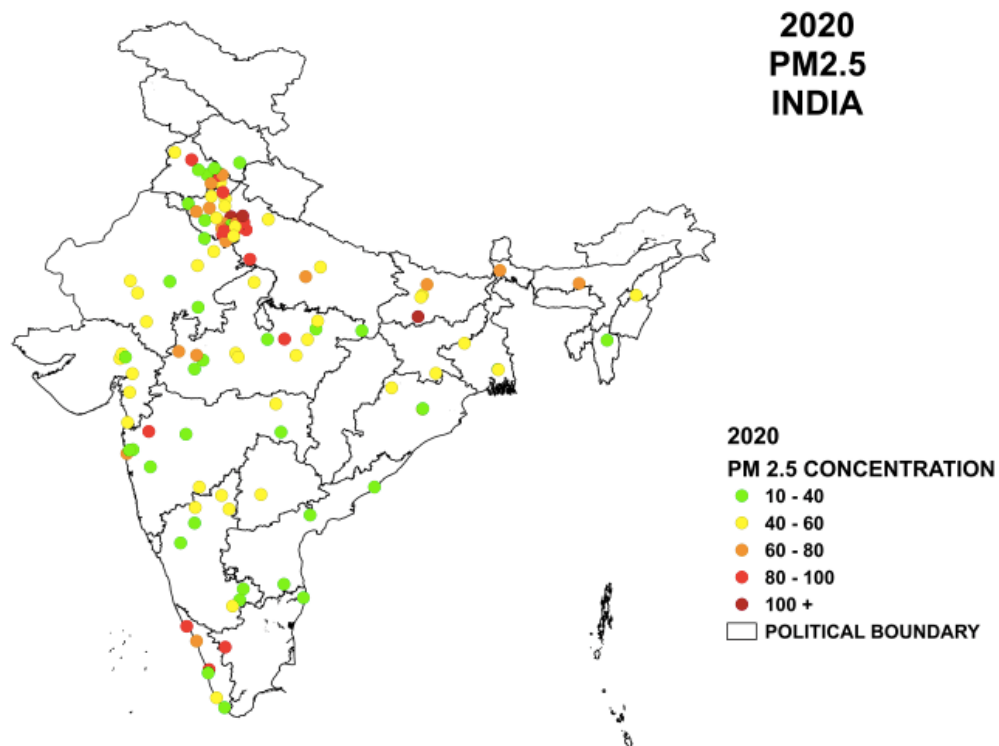
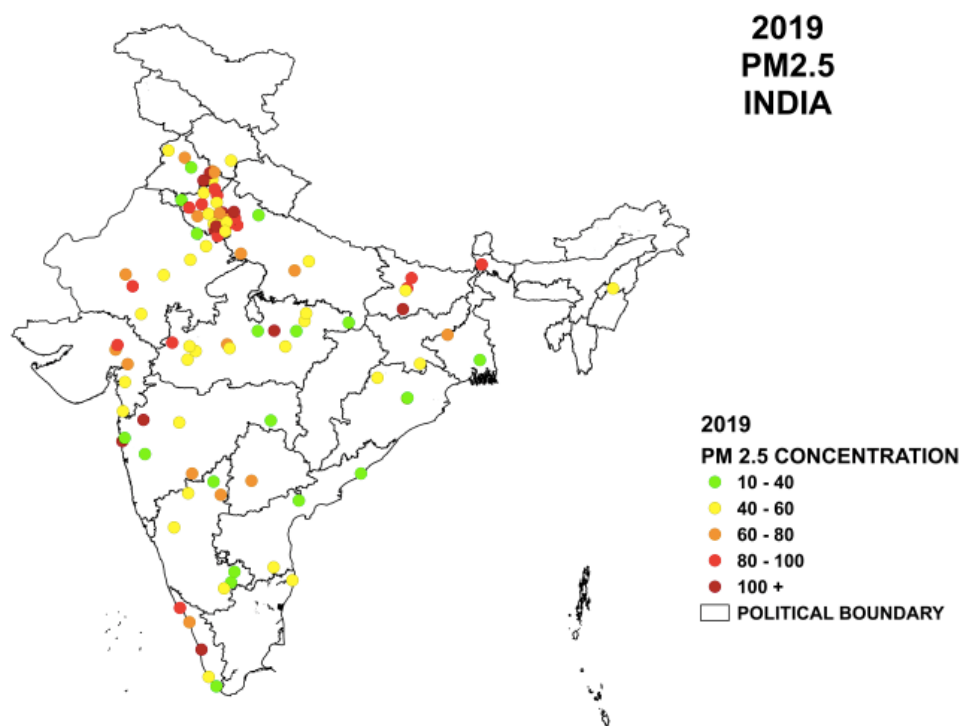
Year-wise PM 2.5 Concentration Maps of Major Indian Cities

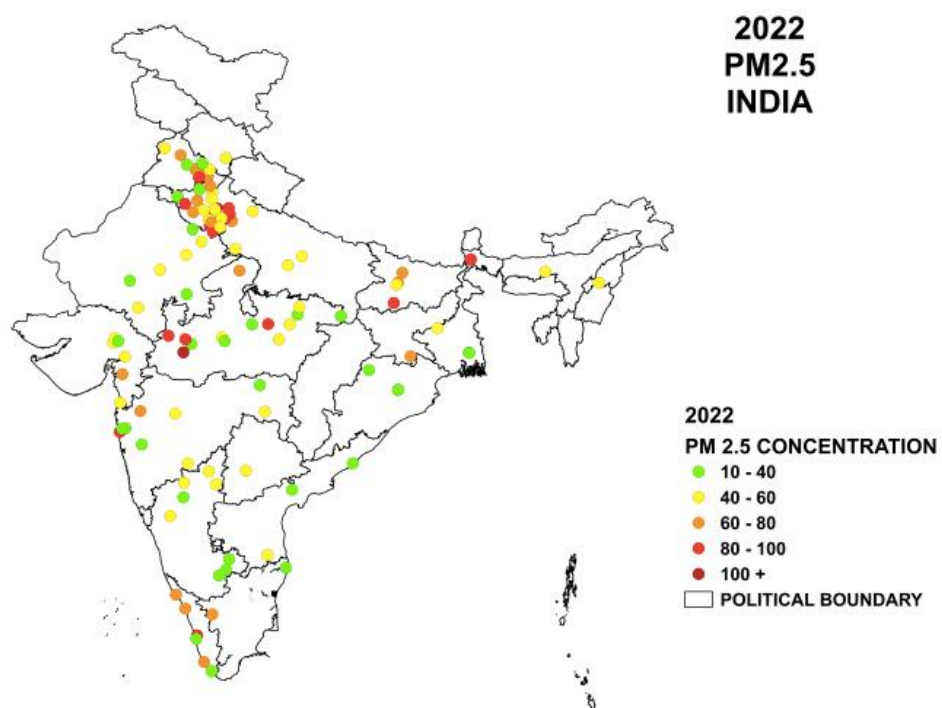
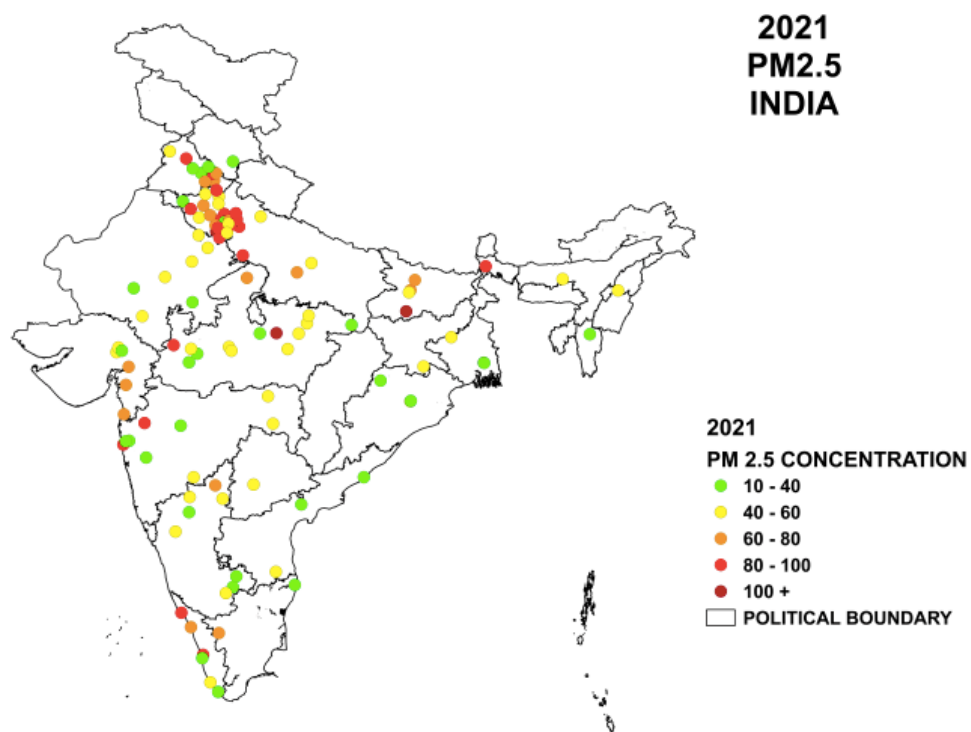
This section presents a detailed **year-by-year visual analysis** of PM 2.5 concentrations across major Indian cities using GIS-based mapping. Each map reflects the annual average levels of PM 2.5 for a specific year, offering a clear understanding of how air pollution levels have changed over time in different parts of the country.

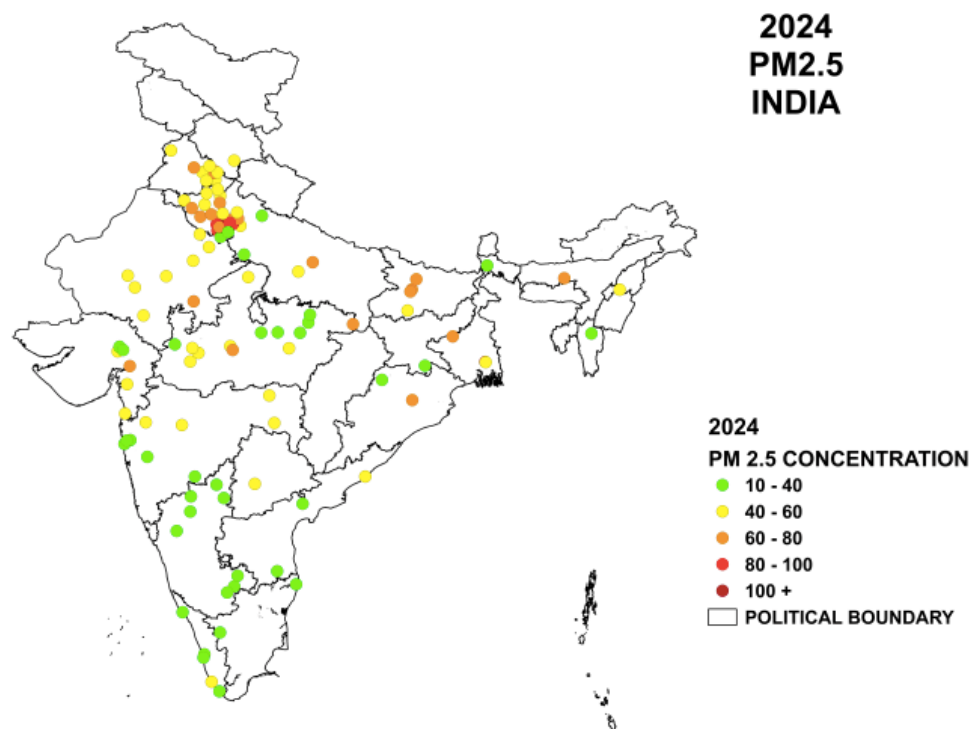
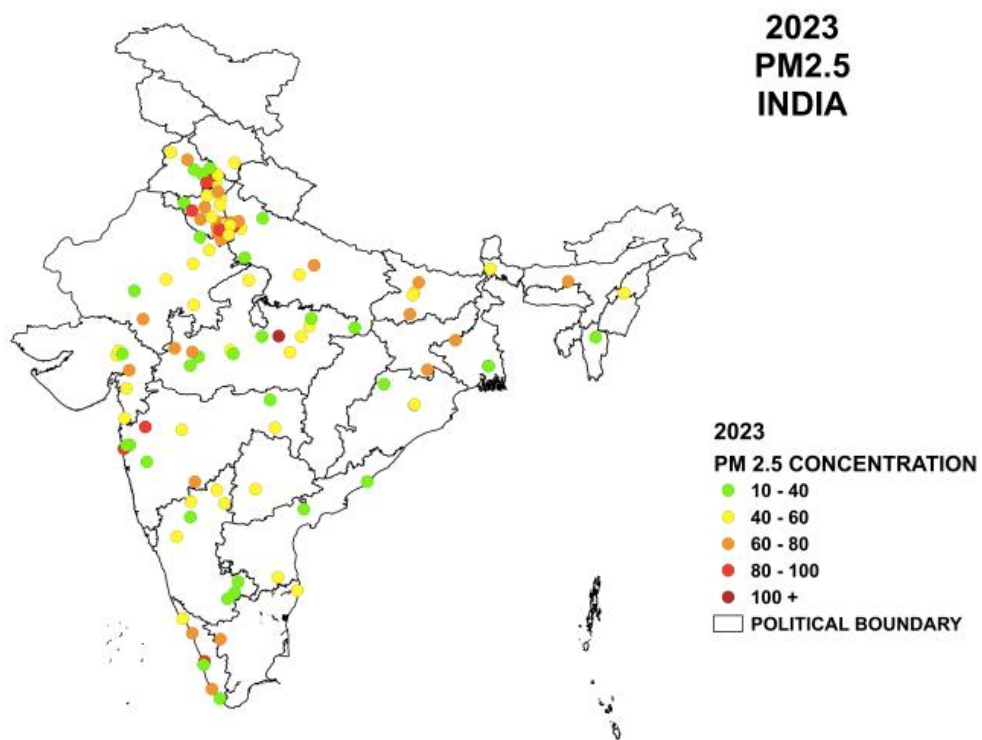
By plotting this data annually from **2018 to 2024**, it becomes possible to observe:

- How specific cities have improved or worsened over time,
- Which regions consistently remain polluted,
- And how external factors (such as the **COVID-19 lockdown in 2020**) influenced nationwide air quality.









4. Annual Trend Analysis

Annual maps were analyzed to observe:

- Year-on-year **changes in pollution levels**
- Cities with **rising or falling trends**
- **Post-COVID impact (2020)** and the recovery pattern in 2021–2023

To assess the long-term trends in PM 2.5 concentrations across Indian cities, the **Mann–Kendall trend test** was applied to annual average data from **2018 to 2024**. This non-parametric statistical test helps detect whether a **monotonic trend** (increasing or decreasing) exists over time, without assuming a specific data distribution.

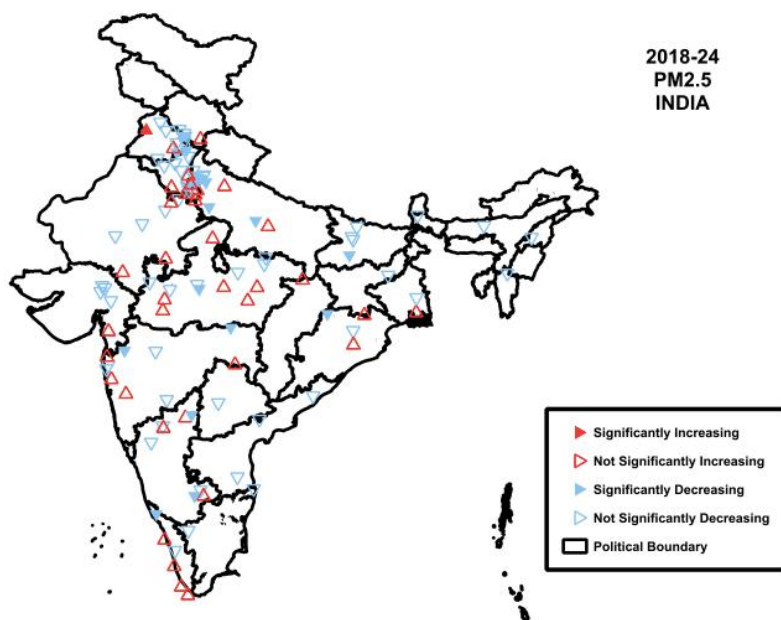
In this analysis:

- The **sign of the slope** indicates the **direction** of the trend (positive for increasing, negative for decreasing).
- The **p-value** determines the **statistical significance** of the trend. A value **less than 0.05** indicates a significant trend; values above suggest the change may not be meaningful.

The results show that while a few cities exhibited significantly increasing or decreasing trends, **most cities fell into the “Not Significantly Decreasing” category**, reflecting either stable PM 2.5 levels or minor fluctuations that were not statistically significant.

This statistical approach adds depth to the GIS-based visual analysis, helping identify cities with clear long-term patterns and those requiring more focused intervention to address stagnant or worsening air quality.

Date Time	Latitude	Longitude	2018	2019	2020	2021	2022	2023	Avg of 5 year	Trends	Computed_S	P_Value	Trend_Category	
Agra	27.17667	78.008072	103.428068	73.530077	80.5061184	80.4437173	57.096586	38.0184297	72.1704994	-10.754602	-10.754602	0.01509605	Significantly Decreasing	
Ahmedabad	23.033863	72.585022	164.119478	61.3986829	47.1944868	52.8495115	45.636997	50.0445904	70.2072911	-17.485842	-17.485842	0.1170274	Not Significantly Decreasing	
Aizawl	23.8	92.9			17.7099474	19.4924928	7.87011218	11.5178442	14.1475991	-3.019869	-3.019869	0.27844576	Not Significantly Decreasing	
Ajmer	26.25	74.66667	58.5423247	47.784905	39.9376093	45.5808997	52.1721697	46.3963702	48.4023798	-1.1978482	-1.1978482	0.49196391	Not Significantly Decreasing	
Alwar	27.5	76.5	51.9819949	45.217469	41.2029606	43.4956289	41.3720788	42.1132826	44.2305691	-1.6739161	-1.6739161	0.0752999	Not Significantly Decreasing	
Amaravati	16.5131	80.5165	39.2138619	39.30413	29.0331509	28.8809354	29.4727534	34.1732735	33.3463509	-1.5671225	-1.5671225	0.21929992	Not Significantly Decreasing	
Ambala	30.37818	76.776695			59.5846055	58.1631382	60.8539218	69.4497101	42.980232	58.2063215	-2.1922175	-2.1922175	0.54961091	Not Significantly Decreasing
Amritsar	31.63398	74.872261	42.3195312	49.9332238	47.2462008	50.8371494	50.3966965	59.7736538	50.0844093	2.63577084	2.63577084	0.02653967	Significantly Increasing	
Ankleshwar	21.63236	72.99001			47.8889796	51.7465891	69.7512736	65.0928634	49.6920923	56.8343596	1.69524996	1.69524996	0.65949866	Not Significantly Increasing
Asansol	23.673944	86.952393	61.6914059	61.3028245	55.6061207	59.2091739	57.934518	62.1810876	59.6541884	-0.1158131	-0.1158131	0.87340324	Not Significantly Decreasing	
Aurangabad	19.88467	75.33986	45.8915374	41.2723211	30.5925861	34.7038356	43.0908658	43.7046701	39.8759694	-0.0390701	-0.0390701	0.98152443	Not Significantly Decreasing	
Bagalkot	16.179199	75.699402			29.8602519	20.6252367	24.3162918	23.0537987	24.4638948	-1.6728304	-1.6728304	0.44767983	Not Significantly Decreasing	
Baghpat	28.95	77.2167	139.352544		101.467823		91.9999779	113.999678		110.764013	-6.0173577	-6.0173577	0.35635135	Not Significantly Decreasing
Bahadurgarh	28.68	76.919998			73.5657696	76.9916729	90.786733	84.0416183	75.4648144	80.1701216	1.0848035	1.0848035	0.69684	Not Significantly Increasing
Ballabgarh	28.341639	77.325348			86.911418	79.0210932	82.6562319	68.5719542	66.1313043	76.6584003	-5.2009366	-5.2009366	0.0292228	Significantly Decreasing
Bengaluru	12.972442	77.580643	32.9812677		36.8825742	28.0794663	31.1828274	35.1750346	30.8646697	32.52764	-0.3600642	-0.3600642	0.68694249	Not Significantly Decreasing
Bhiwadi	28.21024	76.86056	128.426725	102.807247	95.2649737	110.341085	94.5929088	77.2718302	101.450795	-7.5811823	-7.5811823	0.04335611	Significantly Decreasing	
Bhiwani	28.799046	76.133514			74.9210302	39.2725256	51.3423311	75.043529	67.1929085	61.5544649	2.031476	2.031476	0.74226726	Not Significantly Increasing
Bhopal	23.259933	77.412613			68.9008189	41.9666303	45.1126429	47.8336017	47.109799	50.1846986	-3.7715068	-3.7715068	0.32944531	Not Significantly Decreasing
Brajrajnagar	21.816667	83.916667	65.0654977		55.2936904	58.4245901	34.5504981	38.3592936	34.5284118	47.7036636	-6.4960775	-6.4960775	0.0139662	Significantly Decreasing
Bulandshahr	28.406963	77.849831	114.564833	93.1101631	99.2049124	93.2087835	74.1108434	59.6975628	88.9828497	-9.6380125	-9.6380125	0.00683597	Significantly Decreasing	
Chandigarh	30.741482	76.768066			66.1063596	33.1760894	38.5069822	44.0129043	58.8256062	48.1255884	-0.3724692	-0.3724692	0.94599643	Not Significantly Decreasing
Chandrapur	19.970324	79.30336	40.43684		36.764436	30.1816186	43.4625454	47.2816922	46.7511115	40.8130406	2.18297295	2.18297295	0.18483146	Not Significantly Increasing
Chennai	13.067439	80.237617	54.0291687	40.5599059	30.5654002	27.1150746	30.1637667	40.2292944	37.1104351	-2.961089	-2.961089	0.25339668	Not Significantly Decreasing	



Trend Analysis of Annual Average PM 2.5 in Major Indian Cities (2018–2024)

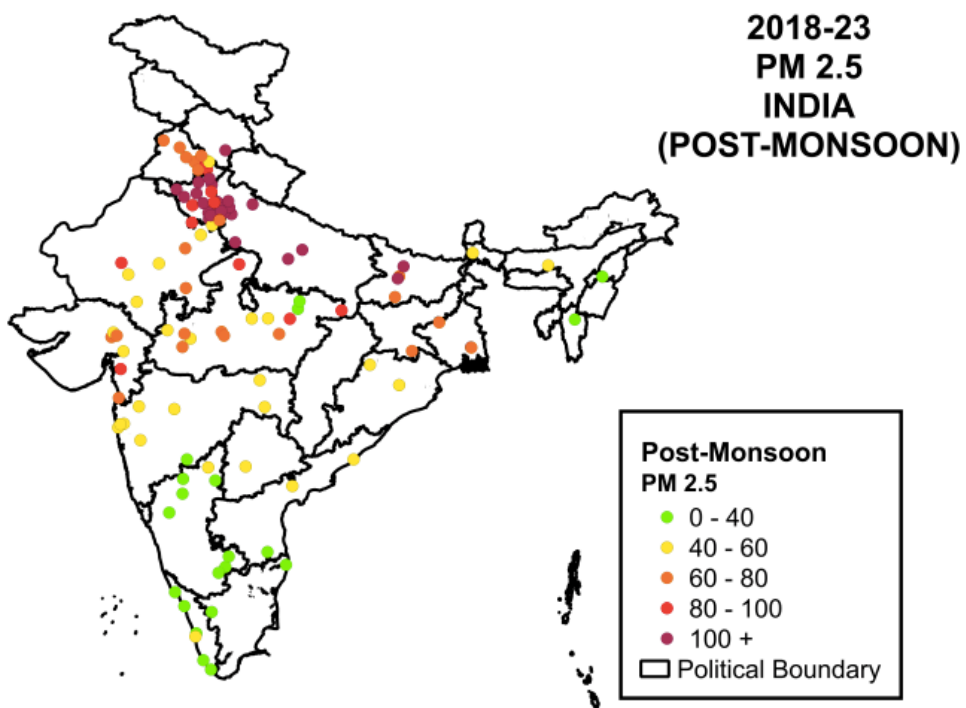
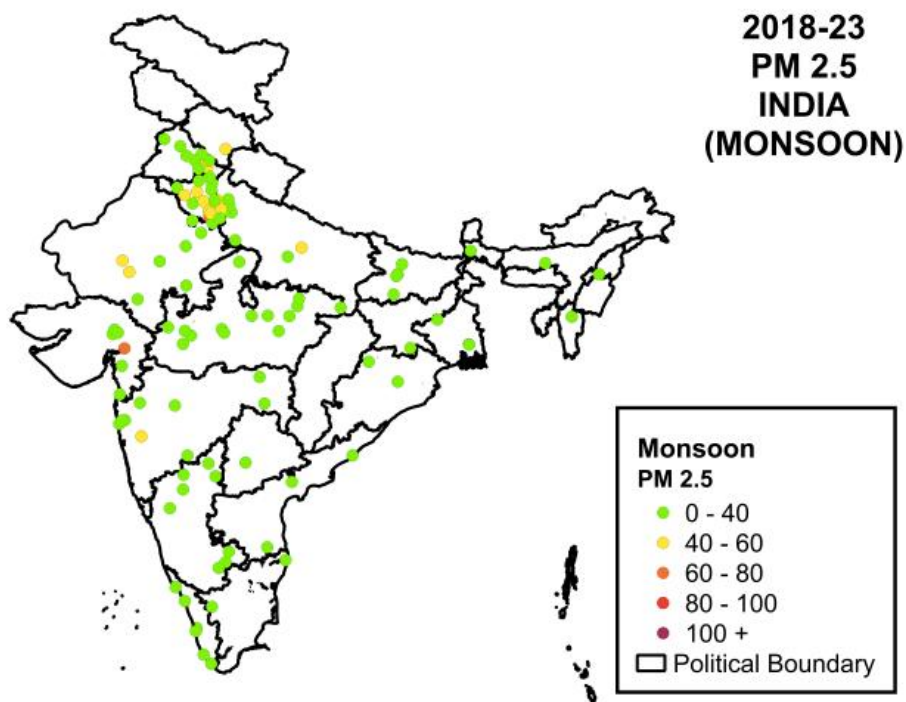
Significantly Decreasing		Decreasing	
Agra	Ahmedabad	Dewas	Maihar
Ballabgarh	Aizawl	Eloor	Mandikhera
Bhiwadi	Ajmer	Fatehabad	Manesar
Brajrajnagar	Alwar	Gandhinagar	Meerut
Bulandshahr	Amaravati	Greater Noida	Mumbai
Dadri	Ambala	Guwahati	Muzaffarpur
Gaya	Asansol	Hajipur	Nandesari
Gobindgarh	Aurangabad	Hapur	Noida
Kaithal	Bagalkot	Hisar	Pali
Kannur	Baghpat	Hubballi	Patiala
Kanpur	Bengaluru	Hyderabad	Patna
Kurukshetra	Bhopal	Jaipur	Ratlam
Mandideep	Chandigarh	Jalandhar	Rohtak
Nagpur	Chennai	Jind	Rupnagar
Nashik	Coalfields	Karnal	Satna
Panchkula	Coimbatore	Kohima	Siliguri
Ramanagara	Damoh	Kolkata	Solapur
Yadgir	Delhi	Ludhiana	Thane
	Tirupati	Vatva	Visakhapatnam

5. Seasonal Categorization

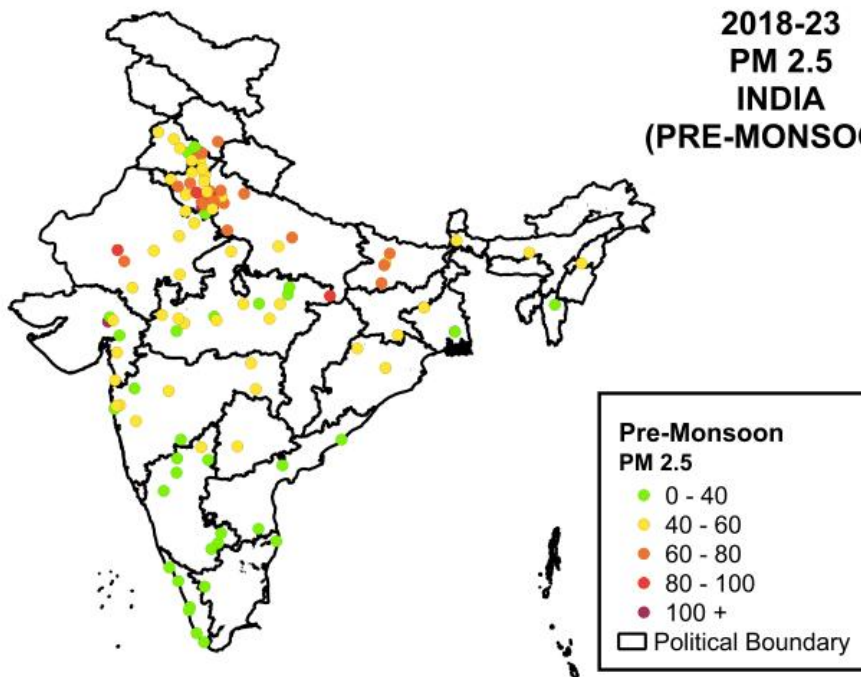
To understand the impact of seasons on air quality, the data was grouped into four distinct seasons:

- **Winter:** December to February
- **Pre-Monsoon:** March to May
- **Monsoon:** June to September
- **Post-Monsoon:** October to November

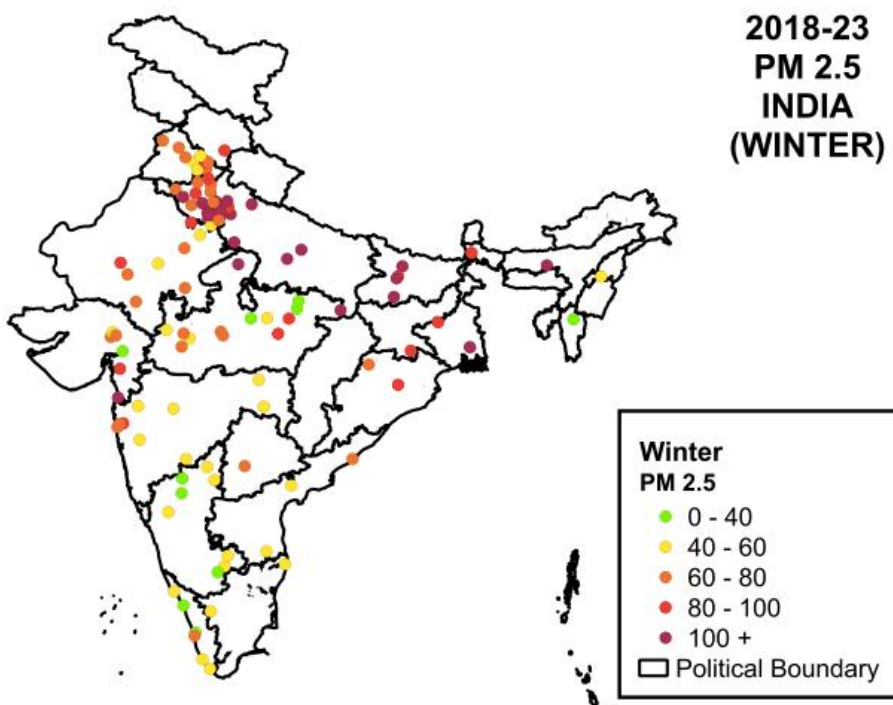
Each city's PM 2.5 values were averaged separately for these periods to identify **seasonal pollution behavior**. seasonal maps were examined to identify the **most and least polluted seasons** and understand how meteorological factors influence PM 2.5 concentrations.



**2018-23
PM 2.5
INDIA
(PRE-MONSOON)**

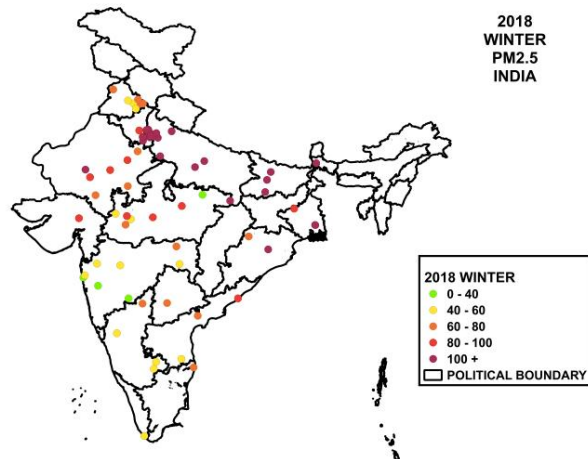
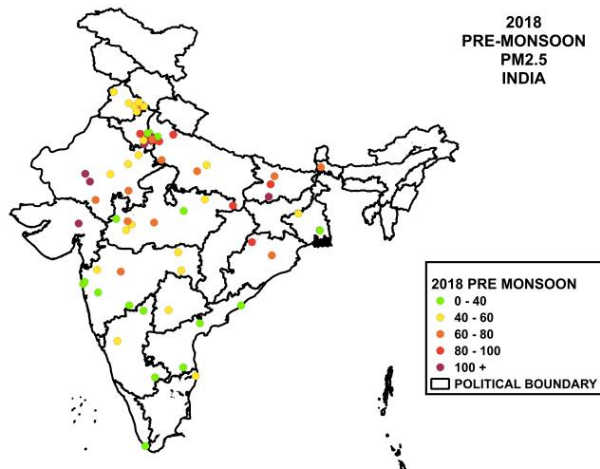
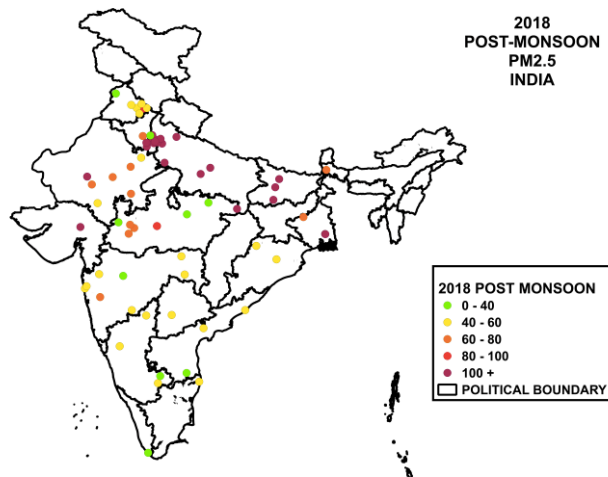
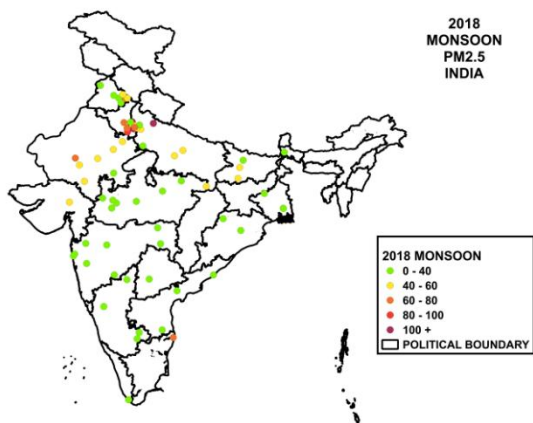


**2018-23
PM 2.5
INDIA
(WINTER)**

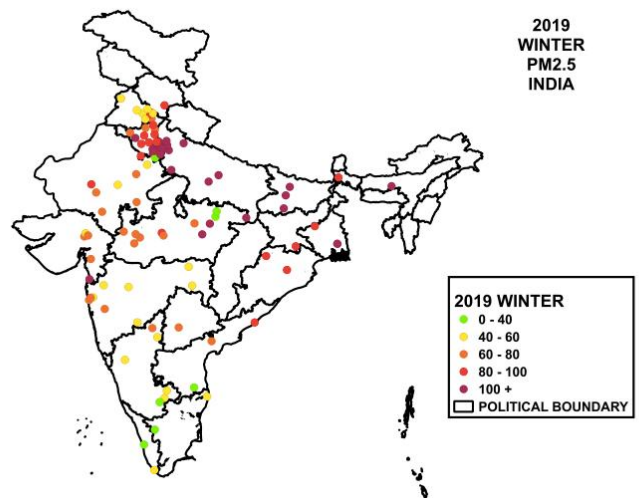
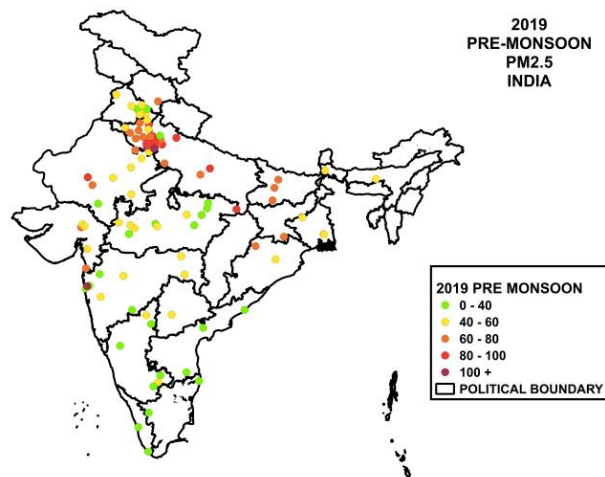
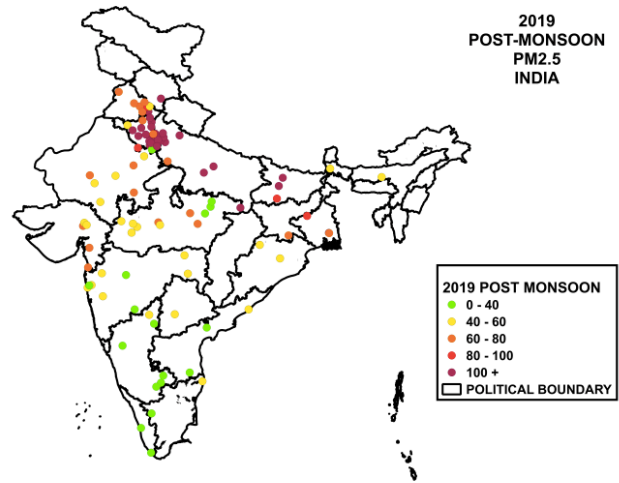
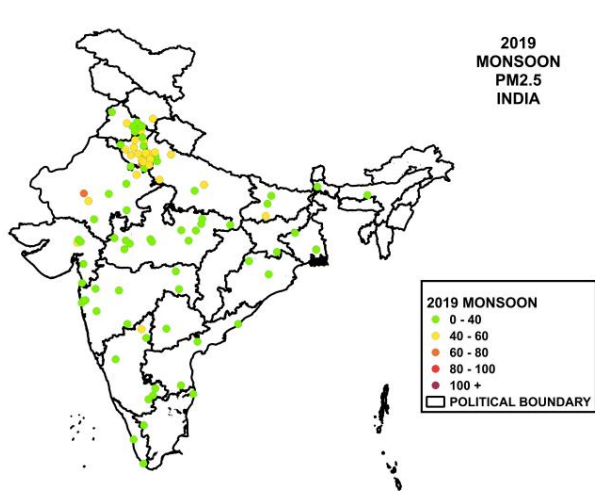


Year-wise Seasonal PM 2.5 Concentration Maps of Major Indian Cities (2018-2024)

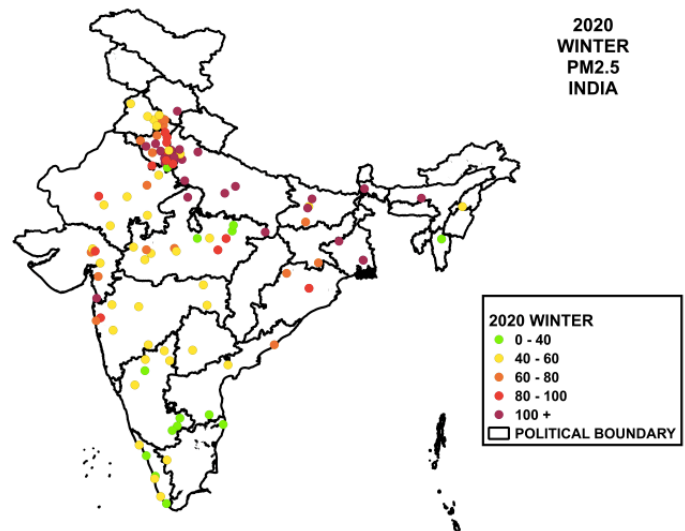
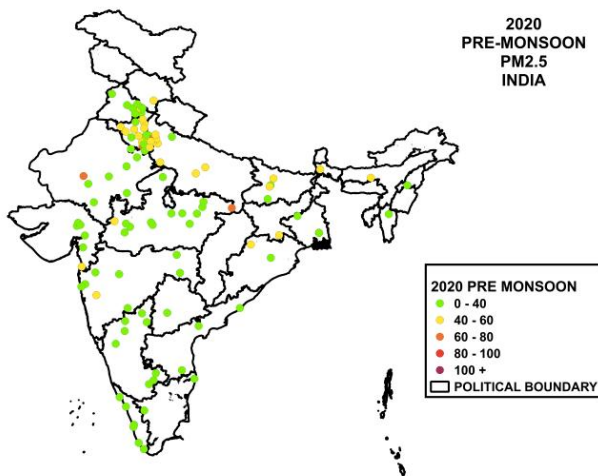
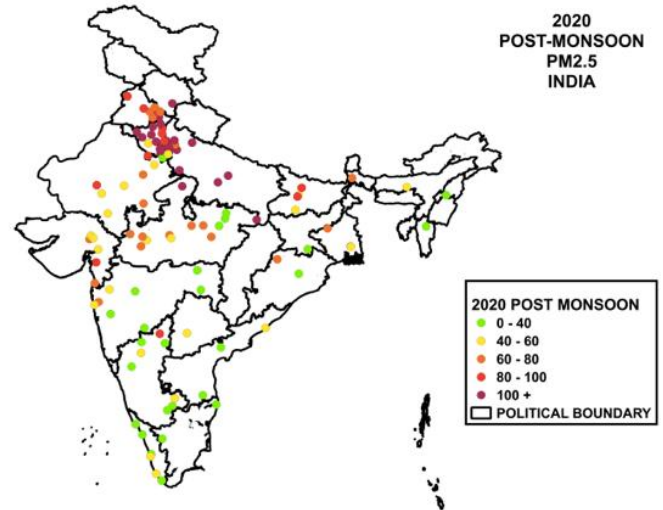
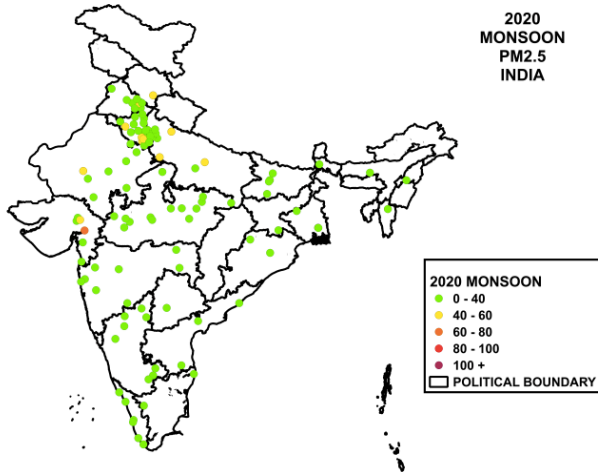
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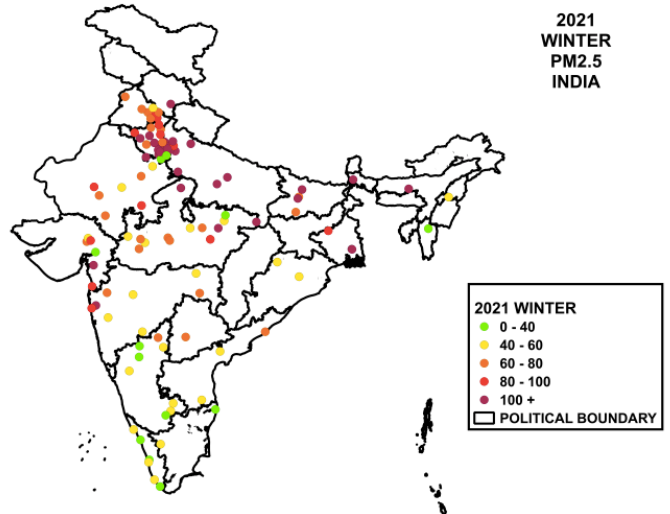
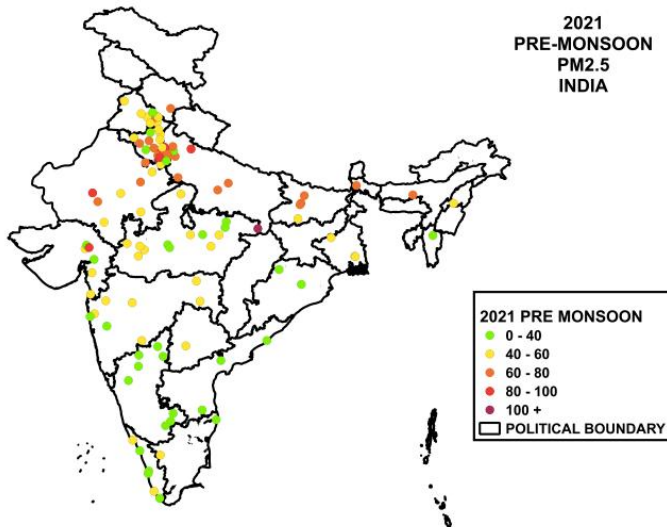
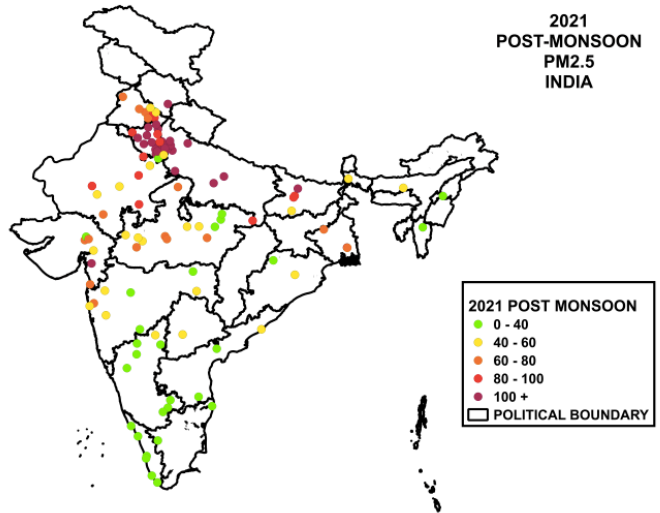
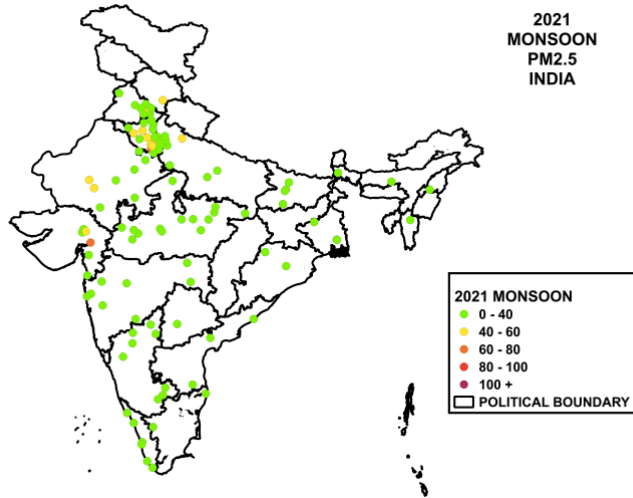
2019



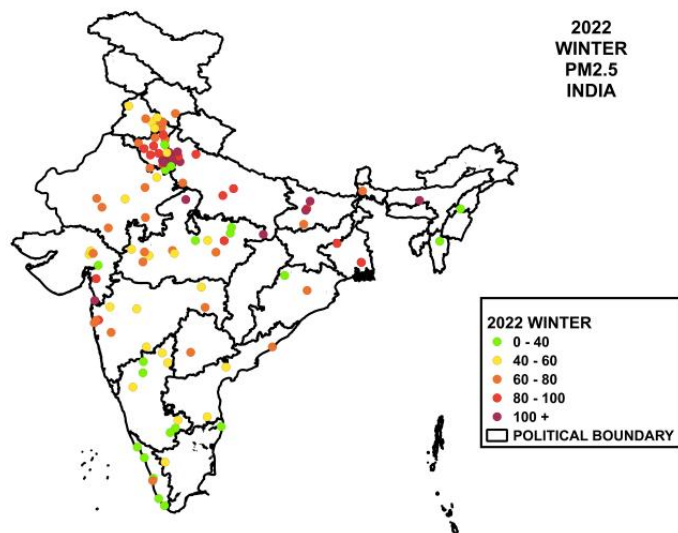
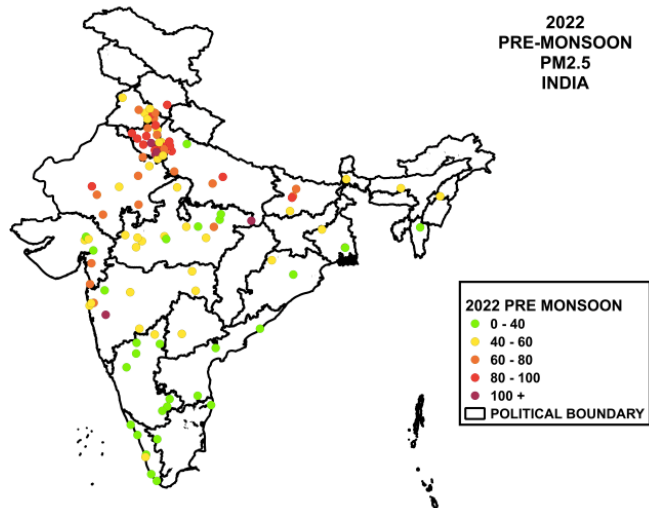
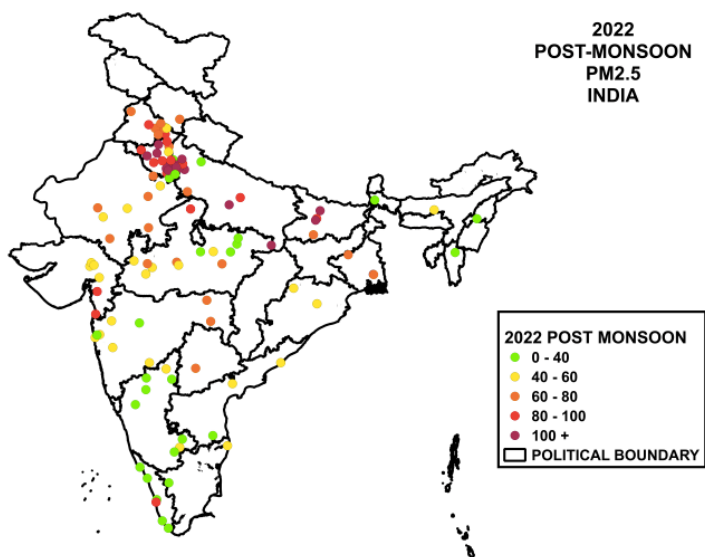
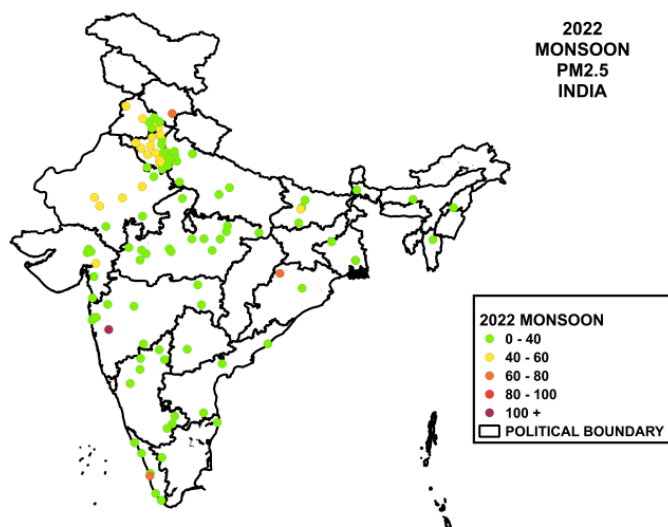
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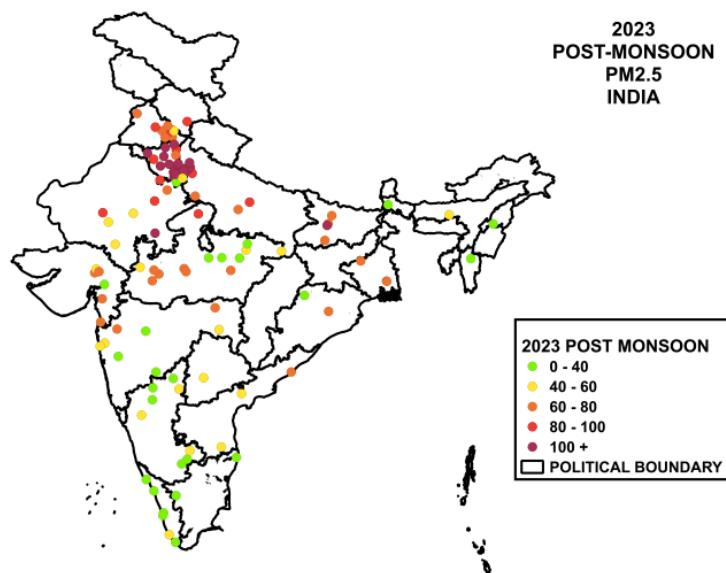
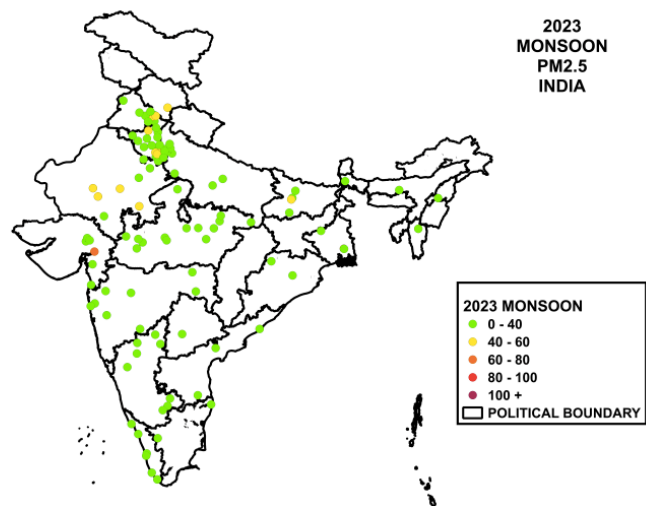
2021



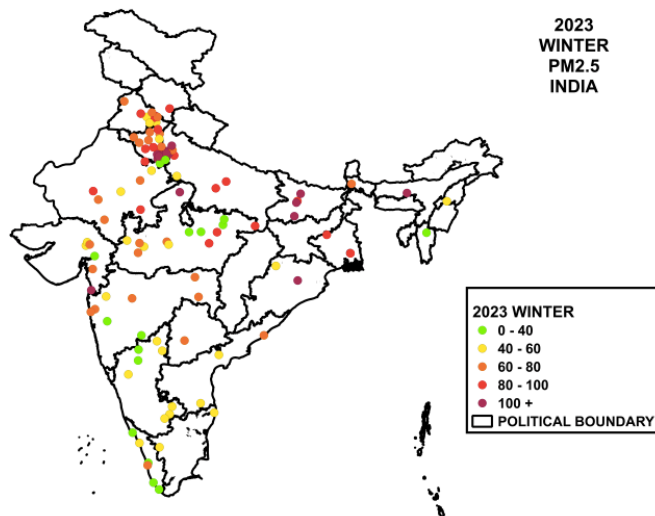
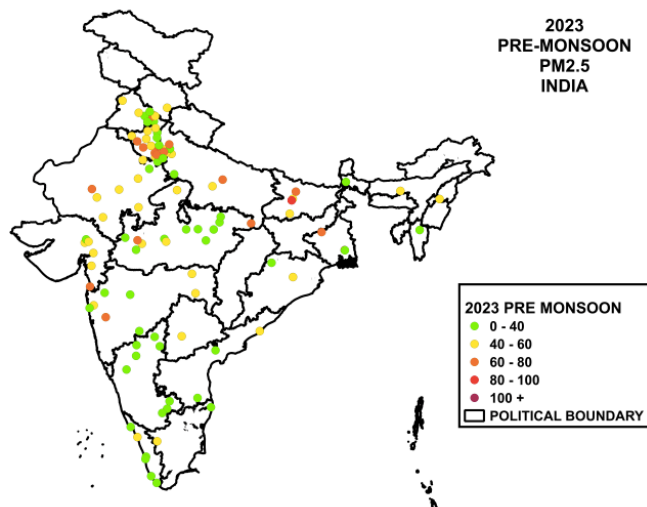
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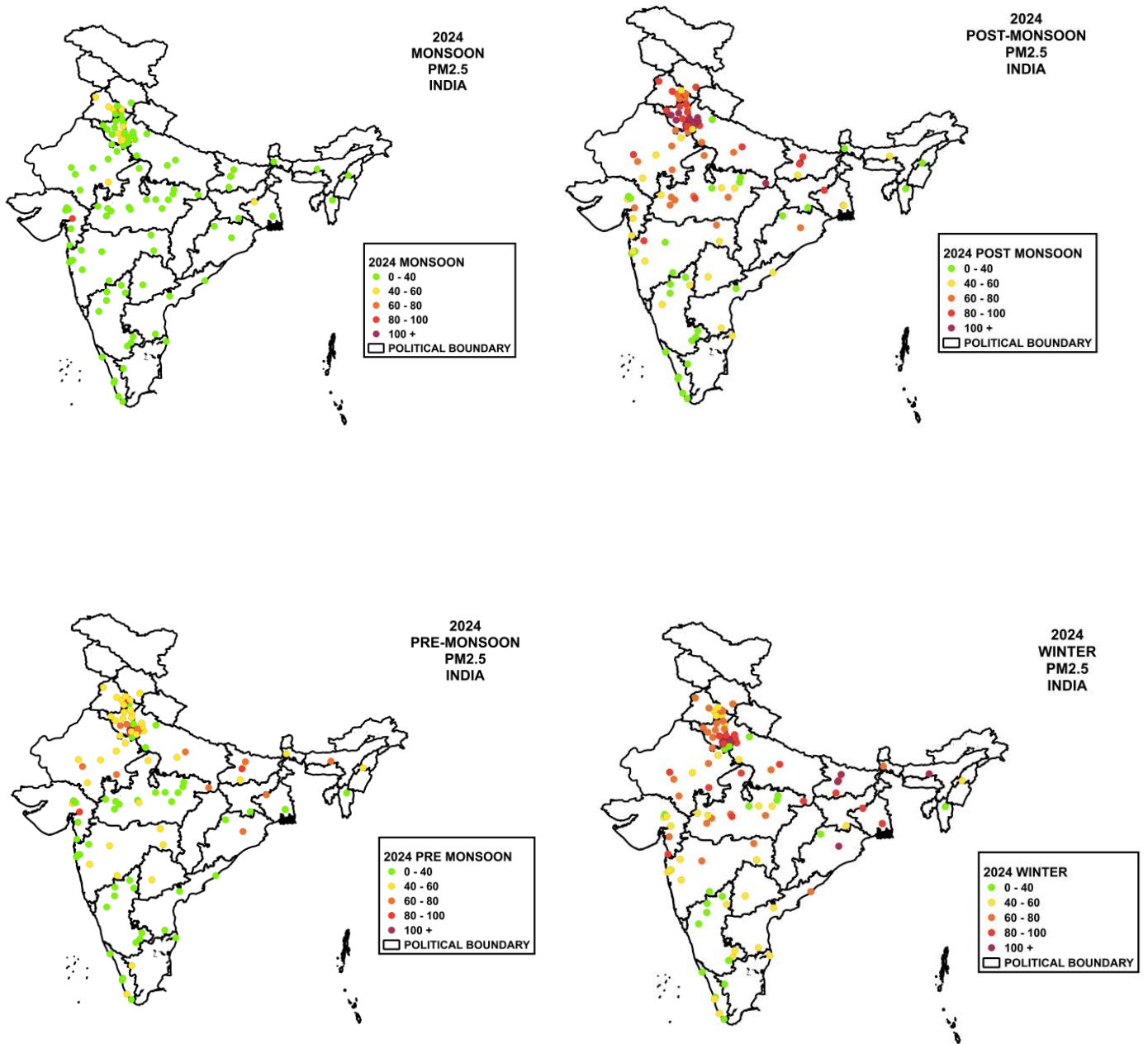
2023



S



2024



Analysis

Regional Analysis

North Zone

- Highest year-round pollution.
- Firecracker ban and GRAP in Delhi showed **limited impact**.
- Needs drastic **policy, public transport, and urban planning reforms**.

South Zone

- Cities like Chennai, Bengaluru, Kochi consistently clean.
- **Low PM 2.5 variation** across seasons.
- Could serve as a **model region**.

West Zone

- Mumbai and Surat show **steady improvements**.
- Dust-prone cities like Ahmedabad fluctuate with **pre-monsoon spikes**.

East & NE Zone

- Kolkata, Patna show increasing post-monsoon PM 2.5.
- NE cities (Shillong, Aizawl) remain **least polluted** but under-monitored.

Year-wise Seasonal Comparison (2018–2023)

Seasonal maps for each year reveal:

- Winters consistently dominate in PM 2.5 intensity.
- **2020 winter** remained cleaner due to post-lockdown inertia.
- Southern cities show **minimal seasonal variation** (almost flat profiles).
- East India (e.g., Kolkata, Guwahati) shows sharp **post-monsoon spikes**.

Seasonal Analysis

Winter:

- Highest concentrations in Delhi, Agra, Kanpur, Patna.
- Contributing factors: **low wind speed, inversion layers, stubble burning, domestic heating.**

Post-Monsoon:

- **Second worst season**, mainly in North & East India.
- Firecracker pollution (Diwali) and residual crop burning add to levels.

Monsoon:

- Lowest PM 2.5 due to **rain washout effect.**
- Cities like Ahmedabad and Hyderabad show near-baseline levels.

Pre-Monsoon:

- Variable results.
- Dust storms in Rajasthan & Gujarat cause **local spikes.**
- North India relatively clean; South remains stable.

Annual Year-Wise PM 2.5 Maps (2018–2023)

2018–2019:

- Baseline years with **consistently high PM 2.5** across North and Central India.
- Sudden peaks during winters and post-monsoon periods.

2020 (Lockdown Year):

- **Sharp drop across all cities.**
- Mumbai, Chennai, and Bengaluru reached decade low levels
- Demonstrated the **impact of anthropogenic activities.**

2021–2024:

- Partial rebound.
- Some cities maintained lower levels post-COVID, suggesting **structural changes in transport and industry**

7-Year Annual PM 2.5 Averages

The combined map showing the 7-year average PM 2.5 levels provides a **nationwide view of chronic exposure**:

- **Top 5 most polluted cities:** Delhi, Kanpur, Lucknow, Patna, Varanasi — all in North India.
- **Top 5 cleanest cities:** Kochi, Bengaluru, Panaji, Chennai, and Thiruvananthapuram.
- **Central belt** (e.g., Raipur, Nagpur, Bhopal): Moderate but rising levels.

The spatial map clearly reveals that **the Indo-Gangetic Plain** is the worst-affected region, where topography and human activity combine to trap pollutants year-round.

City-Wise 7-Year Trend Analysis

GIS maps highlight **pollution trajectories**:

- **Delhi:** Peak average PM 2.5 above ; stagnant despite policy efforts.
- **Mumbai:** Steady decline since 2020
- **Lucknow & Patna:** Sharp increases in recent years; urgent need for control.
- **Bengaluru & Hyderabad:** Consistent low levels with minor fluctuations.
- **Agra, Varanasi, Indore:** Show cyclical patterns; need city-specific analysis.

These trends indicate that **national strategies must be adapted to local pollution patterns, sources, and meteorological influences.**

Key Findings of the Study

The analysis of seasonal and annual PM 2.5 trends across major Indian cities led to several important observations:

Seasonal Variation Dominates Yearly Averages:

The most noticeable shifts in PM 2.5 levels occurred not from year to year, but between seasons. **Winter and post-monsoon months consistently recorded the highest pollution levels**, while monsoon periods showed natural cleansing effects due to rainfall. This highlights the importance of season-specific strategies over general annual plans.

Delhi Remains a Persistent Hotspot:

As expected, Delhi continues to top the list of the most polluted cities. Despite various interventions, it still records some of the highest PM 2.5 levels in winter. However, the worrying trend is that **many Tier-2 cities** like Varanasi, Raipur, and Patna are **silently worsening**, indicating that air quality issues are spreading beyond major metros.

Rainfall as a Temporary Cleanser:

The **monsoon season significantly reduces PM 2.5 levels** across all regions due to the natural washout effect. However, this effect is short-lived. Once the rain stops, especially post-monsoon, pollution levels climb quickly again—suggesting the need for consistent action across seasons.

Pollution Returns Without Continuous Effort:

Even in cities where interventions have shown short-term success, the absence of year-round efforts has led to quick rebounds in pollution levels. This was particularly evident post-COVID, where many cities saw clean air in 2020, only to return to previous pollution levels by 2022.

Benefits of GIS Mapping in PM 2.5 Analysis

The use of **Geographical Information Systems (GIS)** in this project played a central role in transforming raw PM 2.5 data into insightful, visual patterns that could be clearly understood and applied in real-world decision-making.

Clear Visualization:

GIS allowed the creation of heatmaps and thematic layers that made it easy to understand patterns, spot pollution hotspots, and communicate findings visually.

Urban Zoning and Hotspot Detection:

City-level mapping helped identify specific polluted zones, guiding targeted action and policy focus at the micro level.

Tracking Seasonal Pollution Shifts:

GIS revealed how pollution drifts with seasonal winds, topography, and inversion layers, helping understand how PM 2.5 moves over time.

Early Action Planning:

Seasonal maps enabled proactive decision-making—identifying high-risk periods and locations for interventions like cracker bans or dust suppression.

Multi-Layered Analysis:

GIS allowed overlaying of other factors like population, traffic density, and land use, providing a fuller picture of causes and risk areas.

Conclusion

This study offered a comprehensive and data-driven exploration of India's PM 2.5 pollution trends over a seven-year period, using the combined power of **GIS mapping** and **data analytics**. Through the analysis of seasonal and yearly patterns across more than 100 cities, the project shed light on the stark regional disparities and recurring pollution cycles that shape the country's air quality landscape.

One of the most striking observations was the consistent presence of **high PM 2.5 concentrations in North Indian cities**, particularly across the Indo-Gangetic Plain. Cities like Delhi, Lucknow, and Kanpur faced some of the highest year-round exposure, especially during winter and post-monsoon seasons. Meanwhile, **southern and coastal cities** such as Chennai, Bengaluru, and Kochi demonstrated **relatively lower and more stable PM 2.5 levels**, underlining the combined influence of geography, meteorology, and local emission sources.

The use of GIS allowed for not just statistical analysis, but a **visual and spatial understanding** of how PM 2.5 behaves across time and space. Seasonal heatmaps clearly illustrated how **winter and post-monsoon months amplify pollution**, while **monsoon periods offer natural relief** through rainfall-driven cleansing. The maps also revealed emerging hotspots in Central and Eastern India that may require more focused monitoring and early policy intervention.

Notably, the year 2020 served as a natural experiment. The impact of the COVID-19 lockdown resulted in **drastically reduced PM 2.5 levels across most cities**, providing a real-world demonstration of what cleaner air can look like when anthropogenic emissions are curbed—even temporarily.

The insights gained through this project reinforce the need for **season-sensitive, region-specific, and data-informed air quality strategies**. It is evident that India cannot rely on uniform, year-round solutions. Instead, the path forward lies in combining ground-level data, spatial tools like GIS, and public engagement to build targeted, sustainable action plans.

This work not only contributes to a better understanding of air pollution patterns, but also serves as a strong foundation for future environmental research, policymaking, and public health planning in India.